Project

Deliverable 4: Implementation 50 Marks

Refer to the project deliverable for further clarity.

Objective

The goal of this deliverable is to build a fully functional version of the mobile application based on the models, architecture, and designs previously created. This is the most critical and practical phase of the project. Students are expected to demonstrate their development abilities, integrate the system components, and produce an app that responds to user needs and business requirements. The implementation must be aligned with modern development standards, include all required features, and run reliably on mobile devices.

All functionality must be developed using one or more of the approved platforms and technologies. This deliverable is purely practical and forms the basis for later testing, results analysis, and presentation.

What's Required

1. Screen Development

- Create all major UI screens required for the core functionality of the app.
- Each screen must include appropriate interface elements (buttons, text fields, inputs, images, menus, etc.).
- Screens must be fully functional, not just designed.
- Layouts should maintain consistency in styling, layout alignment, font size, padding, spacing, and branding if applicable.
- Ensure that the screen flow is intuitive, accessible, and aligned with the previously submitted wireframes or design plans.

2. Authentication (If Applicable)

- If the system requires user identification, implement a functional authentication mechanism.
- This can include sign-up, login, logout, password reset, or similar features.
- Authentication logic must securely handle user credentials, store tokens or sessions, and restrict access to authenticated users only where appropriate.
- Systems that do not require authentication must still implement some form of user identity logic, if needed for storing user data.

3. Backend Integration

- The app must connect with a functioning backend to store and retrieve persistent data.
- Integration with a cloud database or a local database is required, depending on the nature of the system.
- Backend logic must support full data transactions (create, read, update, delete).
- Data storage should be secure, efficient, and well-structured, in line with the data design in Deliverable 1.

4. Core Functionality (CRUD Operations)

- Implement the application's main features. These should include at least two complete functional modules.
- Each module must support basic data operations: creating, reading, updating, and deleting data.
- Modules should include all necessary screens, logic, and connections to backend systems.
- The modules must reflect the original requirements and use cases.

5. Navigation and Routing

- Implement logical and seamless screen-to-screen navigation.
- Navigation should be context-aware, intuitive, and responsive.
- Ensure stack or fragment-based navigation is used properly to avoid navigation errors or unexpected behaviour.
- The back-stack should work as expected and not crash the app.

6. UI Feedback

- Provide appropriate feedback to the user when they interact with the system.
- Include mechanisms like success/failure messages, input validation errors, notifications, dialogues, or loading indicators.
- Input fields must validate data entry for correctness and completeness.
- All actions should result in visible feedback to guide the user experience.

7. Code Quality

- The code must be structured logically and separated into reusable components, classes, or modules.
- The implementation must follow a consistent style guide.
- Important parts of the code must be documented with clear comments to describe their function and logic.
- Remove unused code and redundant files before submission.
- The codebase must be maintainable and readable.

Marking Rubric – 50 Marks

Criteria	Mark Allocated
System design	15
Prototype Development	20
Prototype integration	15
Total	**50**

End of Deliverable 4